

Relation-level MR evidence

Representation and preprocessing
node permutation; face order; mesh encoding
possible adapter or graph encoding issue

Interpretation is evidence-bounded:
a violation is not automatically a program fault.

Physical-model relations
mirror-y; continuity; Reynolds-Strouhal
possible physical constraint or tolerance issue

Temporal rollout relations
determinism; prefix consistency; horizon convention
possible rollout implementation issue

Cross-SUT method comparison
matched variables; units; meshes; horizons
possible comparability failure

Implementation and runtime

runner; parser;
checkpoint; seed;
environment
possible reproducibility
barrier